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Preparation, characterization, mechanical and biodegradation behavior of polypropylene - chitosan/ZnO nanocomposite

Mohamed S. A. Darwish^a, Mohamed H. Mostafa^a, Labiba I. Hussein^a, Abdaleem H. Abdaleem^b, and Moataz A. Elsayy^a

^aPetrochemicals department, Egyptian Petroleum Research Institute, Nasr City Cairo, Egypt; ^bChemistry department, Menofia University, Menofia, Egypt

ABSTRACT

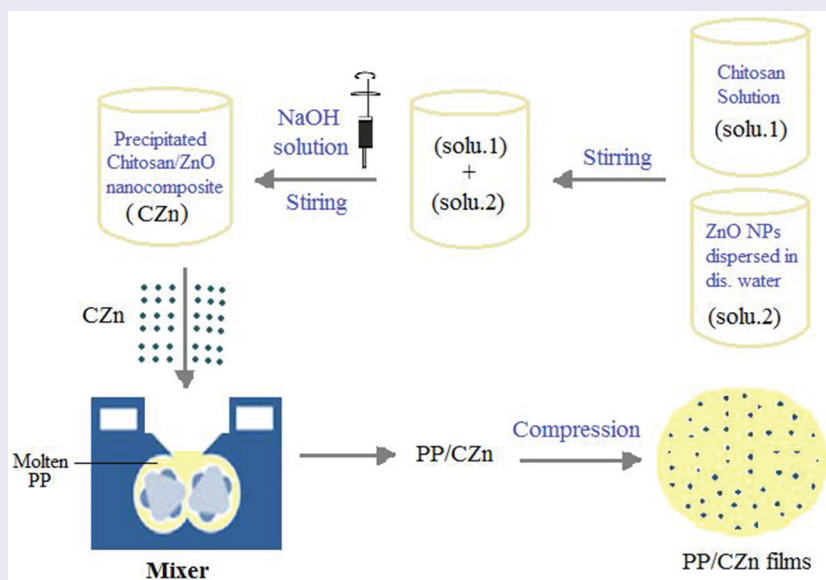
One of the main challenges in the applications related to the environment is improving the biodegradability of polypropylene (PP). The incorporation of filler based on chitosan/ZnO nanocomposite (CZn), prepared via solution mixing technique at room temperature, into PP matrix holds promise in solving this issue. The tensile tests indicated the improved mechanical properties of PP/1CZn compared to PP, however increasing CZn content resulted in deteriorated mechanical properties. The soil burial test results after 3 months indicated the improved biodegradability of PP/CZn relative to PP and the significant impact of UV pretreatment on biodegradation rate.

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KEYWORDS

Polypropylene; chitosan; ZnO; biodegradability; mechanical properties



1. Introduction

Polypropylene (PP) is a polyolefin type which is increasingly produced and used in the world. PP reveals interesting characteristics like ease of processing, outstanding mechanical and thermal behavior, wide availability as well as low cost of production.^[1] PP is utilized in many fields such as medicine, packaging, transportation sector, storage boxes and household appliances.^[2–5] However, due to the increased use of PP in many applications, huge quantities of PP wastes has accumulated in

the environment as a result of its non-biodegradability.^[6] This disadvantage stems from the high molecular weight of PP, long chains of carbon atoms which give resistivity against hydrolysis, hydrophobic surface of PP films, PP crystallinity and the absence of active functional groups. These factors restrict the degradation of PP by microorganisms.^[7,8] The design of PP, possessing acceptable degree of biodegradability, has become an essential issue for reducing the environmental impact of PP wastes. Several strategies have been carried out for converting PP into eco- friendly degradable polymer.^[9]

For example, pro-oxidant addition to PP,^[10] blending with biodegradable polymers,^[11] grafting with hydrophilic monomers^[12] and chemical/physical pretreatments.^[13,14]

Amongst all attempts to improve PP biodegradability, blending with biodegradable polymer has been increasingly reported.^[15,16] PP blends with cassava starch have been prepared with improved biodegradability, reported by Henry C. O. et al.^[17] The biodegradation rate of PP films loaded with 10% starch was 0.015 g/day in sea water environment as reported by Uyigue L.^[18] PP did not reveal water absorption or weight loss during soil burial. However, after soil burial and soaking in water, PP films with 10% and 20% of starch showed changes in their weight.^[19] A study by Ming M. P. et al. showed that PP/starch blend has revealed deterioration in tensile properties and weight loss after the outdoor soil burial.^[20] As studied by Sanjay K. C. et al, PP blends with banana fiber, bamboo fiber as well as pineapple leaf fiber, based on fiber content, have exhibited partial biodegradation (5–15%) in 70–90 days.^[21] A report by N. Ploypetchara et al indicated that the increased content of polylactic acid in PP films, using PP grafted maleic anhydride as compatibilizer, has resulted in decreased crystallinity, melting temperature and elongation at break. While loading PP films with polylactic acid resulted in increased tensile strength, modulus, biodegradation rate and water vapor permeability of PP.^[22]

Amongst several biodegradable polymers, chitosan (Cs) has been of great interest attributed to its non-toxicity, biodegradability, wide availability and biocompatibility. Cs, the deacetylated form of chitin, is composed of linked D-glucosamine and N-acetyl-D-glucosamine units through β - 1, 4 glycosidic linkages.^[23–25] PP blending with Cs is expected to result in some drawbacks with regard to mechanical properties due to

the poor mechanical properties of Cs compared with PP.^[26,27] On this point, incorporating ZnO nanoparticles (ZnO NPs) with remarkable physical properties as reinforcement nanofiller^[28,29] into Cs matrix can act as appropriate solution for reinforcing PP blend with Cs. ZnO NPs is one of the most utilized nanoparticles owing to its biocompatibility, availability and low production cost.^[30] Therefore, this study aims to physically incorporate ZnO NPs into Cs matrix prior to the incorporation into PP matrix and investigate the mechanical and biodegradation behavior of the prepared material.

2. Materials and experiments

2.1. Materials

Polypropylene was obtained from ExxonMobil. Chitosan (low molecular weight, extracted from crab shells, degree of deacetylation: 85%) was obtained from Marine chemicals, India. ZnO nanopowder and glacial acetic acid were obtained from Sigma Aldrich, USA. All solutions were prepared using distilled water.

2.2. Physical preparation of chitosan/ZnO nanocomposite (CZn)

Chitosan/ZnO nanocomposite has been physically prepared by solution mixing method at room temperature, denoted as CZn (Figure 1). 0.5 g of Cs was dissolved in 100 ml of 1% acetic acid solution (solution 1). 0.5 of ZnO NPs was dispersed in 100 ml of distilled water (solution 2). The mixture solutions of 1 and 2 were stirred in quick fit bottomed round flask for 30 minutes at room temperature. The ammonium hydroxide solution was taken in a syringe and added drop by drop to the mixture solution. The precipitated product was taken out of the solution and dried at room temperature.

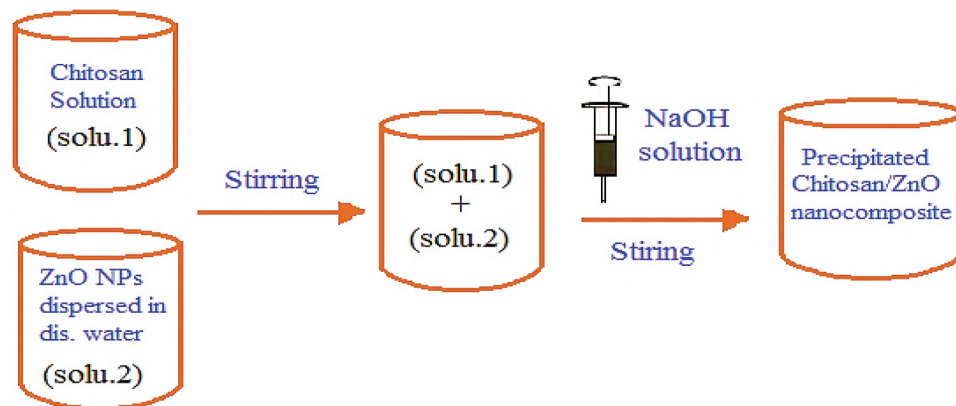


Figure 1. Graphical representation for preparation of chitosan/ZnO nanocomposite by solution mixing technique.

2.3. Preparation of polypropylene – chitosan/ZnO nanocomposites (PP/CZn)

PP and its nanocomposites with 1, 5, 10 and 20% of CZn were prepared from melt by melt processing technique using barbender mixer, denoted as PP, PP/1CZn, PP/5CZn, PP/10CZn and PP/20CZn, respectively (Figure 2). The mixing was achieved by using 60 rpm and 180 °C as rotor speed and temperature, respectively. First, PP was melted for 5 min. After that, various amounts (1, 5, 10 and 20 wt. %) of CZn were added in 3 min time intervals for each. The mixing was continued for another 3 min. The obtained nanocomposites were allowed to cool at ambient temperature. After cooling, PP and its nanocomposites films were prepared via compressing nanocomposites using an electrically heated hydraulic press at 180 °C under a pressure of 200 kg/cm² for 3 min. The average thickness of films was 0.14 ± 0.01 mm. Figure 3 shows the graphical representation of the preparation process. After pressing, the samples were allowed to cool at room

temperature. The films were cut according to standard specifications for other tests.

2.4. Characterization techniques

2.4.1. Transmission electron microscopy (TEM)

TEM-1230 with an accelerating voltage of 100 KV (JEOL Co., Japan) has been used to obtain TEM images of sample.

2.4.2. Fourier transform infrared spectroscopy (FT-IR)

Via using Shimadzu FTIR – 4200 spectrophotometer, the FT-IR spectra of samples in the range (400–4000 cm⁻¹) were obtained.

2.4.3. X-ray diffraction spectroscopy (XRD)

Through using modern PAN analytical diffractometer, Xpert PRO model, the XRD patterns of samples were obtained. Cu K-Alpha radiation (1.54060 Å), 2θ ranging from 4.01° to 69.99°, scanning rate of 0.02°/step and constant operating conditions (40 KV & 40 mA).

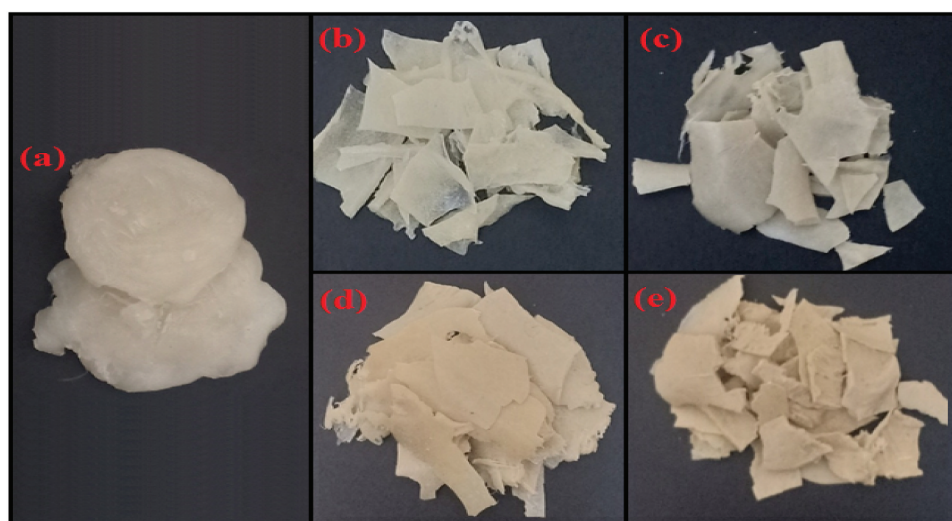


Figure 2. Photos of PP (a), PP/1CZn (b), PP/5CZn (c) PP/10CZn (d) and PP/20CZn.

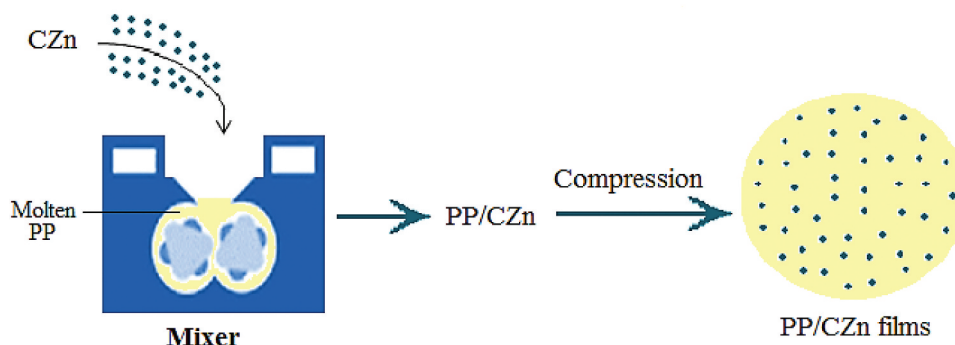


Figure 3. Graphical representation for preparation of PP/CZn film.

2.4.4. UV-vis spectroscopy

V-750 UV-Vis spectrophotometer has been used to obtain the UV-Vis absorption spectra of ZnO NPs, Cs and CZn in the wavelength range of 300–600 nm.

2.4.5. TGA analysis

SDT Q600 V20.9 Build 20 thermal gravimetric was used for the thermogravimetric analysis of samples.

2.4.6. Mechanical properties

A universal testing machine (Zwick/Roell/Z010) was used to carry out the tensile test of samples with regard to standard test method (ASTM D 412).

2.4.7. Biodegradation studies

The soil burial test was used to study the biodegradation behavior of PP and PP/CZn sample films. The films were buried in active soil garden for 3 months to evaluate weight loss percentages of films after burial in soil. The films were cut into (2.0 cm x 1.0 cm) strips and buried in soil at pH 7.5–8.5, temperature 30–35°C and moisture content of 25–30%. The samples were collected distinct time intervals, brushed, washed many times with distilled water and dried to constant weight under vacuum. The film biodegradation was evaluated in the matter of weight loss percentages via gravimetric measurement. Also, the UV pretreatment influence on weight loss percentages of films was studied via pretreatment of films with UV radiation for 72 hours prior to their burial in soil. Also, FT-IR was utilized to follow up the biodegradation process by monitoring changes in the chemical structures of samples before and after soil burial for three months.

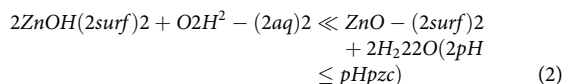
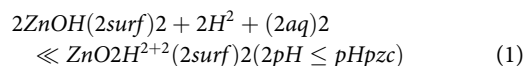
3. Results and discussions

3.1. Preparation and characterization of chitosan/ZnO nanocomposite (CZn)

3.1.1. Mechanism of preparation

The contact between ZnO NPs and water results in forming monolayer of surface hydroxyl groups. Based on solution pH, these groups are either deprotonated (when they release protons) or protonated (when they receive protons). This allows ZnO NPs to exhibit electrical charges on their surface, relying on pH of solution as well as their point of zero charge (PZC). The pH at zero point of charge (pHpzc) is the pH at which the sum of all positive charges on surface balances sum of all surface-negative charges. At solution pH value lower than that of pHpzc, ZnO NPs exhibit surface positive charge, while at solution pH value

higher than pHpzc, ZnO NPs surface is negatively charged.^[31] This is described by equations, (1) and (2).



Regarding the physical preparation of CZn, an appropriate pH value below pHpzc of ZnO NPs and pKa of Cs would allow both ZnO NPs surface and Cs molecular chains to possess positive charges (Figure 4). As a result of the electrostatic repulsion forces between Cs matrix and ZnO NPs, the flexible chains of Cs molecules could be stretched out, leading to dispersing ZnO NPs into Cs matrix.^[32]

3.1.2. Morphology

TEM was used to investigate the morphology of CZn as presented in Figure 5. It was indicated that ZnO NPs are dispersed in Cs matrix, showing the successful preparation of CZn.

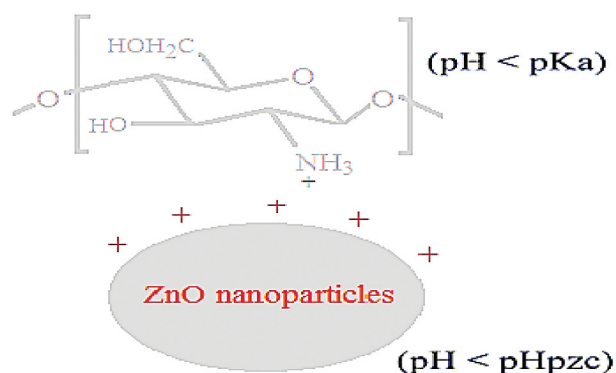


Figure 4. Electrostatic repulsion between positively charged ZnO NPs and Cs chains.

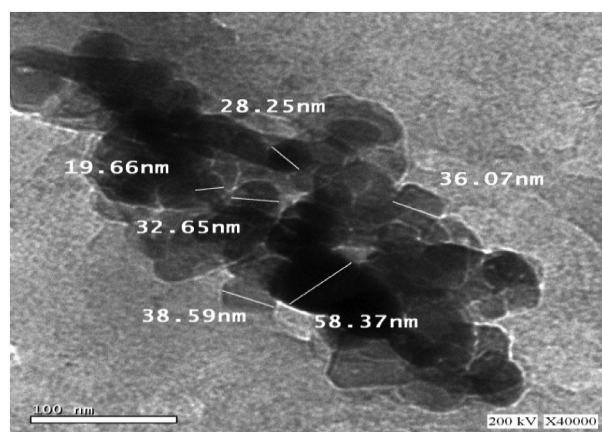


Figure 5. TEM images of CZn.

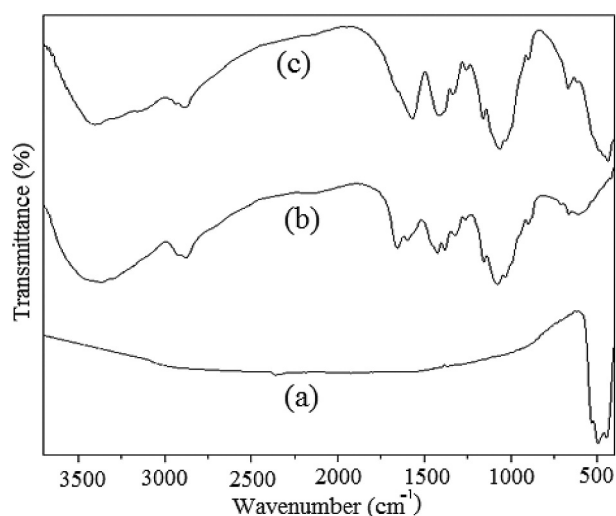


Figure 6. FT-IR spectra of ZnO NPs (a), Cs (b) and CZn (c).

3.1.3. FT-IR spectroscopy

Figure 6 shows the FT-IR spectra of ZnO NPs, Cs and CZn. Figure 6a indicates that ZnO NPs possessed band at 451 cm^{-1} which corresponds to stretching vibration of Zn-O. Figure 6b shows that Cs exhibits bands at 3358, 2869 and 1662 cm^{-1} which were owing to stretching vibrations of ($-\text{NH}_2$ and $-\text{OH}$), $-\text{CH}$ and $-\text{C}=\text{O}$ (amide I), respectively. The peaks at 1590, 1376 and 1149 cm^{-1} are due to N-H bending (amide II), stretching of alcoholic group ($\text{CH}_2\text{-OH}$), and (1-4) glycosidic bond in polysaccharide unit. The band at 1078 cm^{-1} was assigned to C-O-C stretching vibration.^[33] The FT-IR of CZn, presented in Figure 6c, shows that CZn reveals the characteristic peaks of Cs and ZnO NPs. While, the peaks assigned to stretching vibration of amine and hydroxyl groups exhibit shift to higher wave number at 3401 cm^{-1} , which was due to the miscibility between Cs functional groups and those present on ZnO NPs surface.^[32] Also, this may be attributed to the formation

of hydrogen bonds between functional groups present in Cs and on ZnO NPs surface.^[34,35]

3.1.4. XRD spectrometry

The XRD patterns of Cs, ZnO NPs and CZn are displayed in Figure 7. As presented in Figure 7a, Cs demonstrates broad peak at 20.43° , showing that Cs possesses semicrystalline nature.^[36] From Figure 7b, ZnO NPs exhibit sharp peaks at 31.82° , 34.39° , 36.31° , 47.61° , 56.59° , 62.88° , 66.39° , 68.12° , 69.14° and 77.05° . These data are in accordance with standard data of ZnO hexagonal phase (JCPDS card no. 36-1451).^[37] The XRD pattern of CZn (Figure 7c) reveals that CZn displays characteristic peaks of Cs and ZnO NPs, but with changes in the intensities and sharpness. This indicates that both materials affect crystallinity of each other. It can be indicated that ZnO NPs possessed significant impact on Cs in CZn which may be owing to hydrogen bond formation between Cs functional groups and those present on ZnO NPs.

3.1.5. UV-visible spectroscopy

Figure 8 displays the UV-Vis absorbance spectra of ZnO NPs, Cs and CZn. ZnO NPs possess wide band gap energy and the transfer of electron from valence band to conductive band demonstrates remarkable absorption peak. The absorption peak of ZnO NPs was found to be at 373 nm, while CZn revealed slight shift in the absorption peak and appeared at 379 nm. This could be attributed to the good dispersion of ZnO NPs into Cs matrix.^[38]

3.1.6. TGA analysis

TGA was used to study the thermal behavior of Cs and CZn as showed in Figure (9). The decomposition of CZn and Cs at 100°C was due to water removal. CZn exhibits on set decomposition temperature at 244°C which is

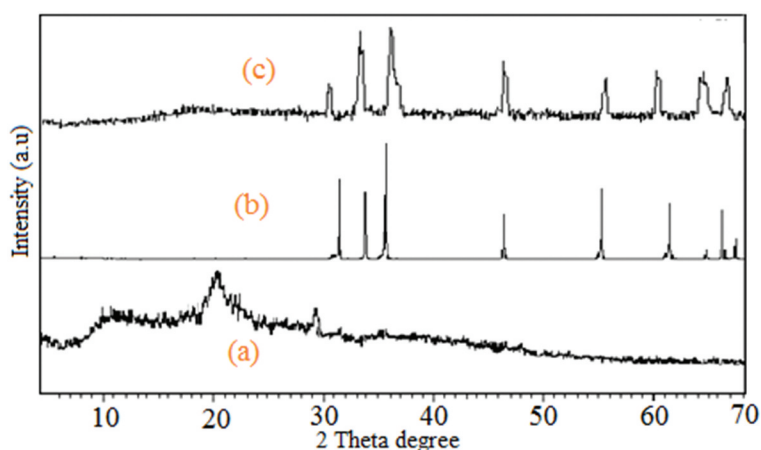


Figure 7. XRD patterns of Cs (a), ZnO NPs (b) and CZn (c).

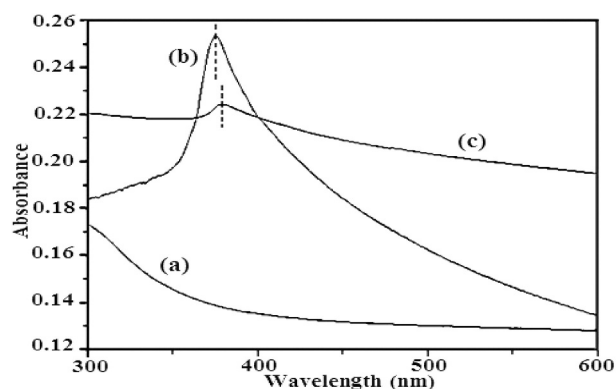


Figure 8. UV-Visible spectra of Cs (a), ZnO NPs (b) and CZn (c).

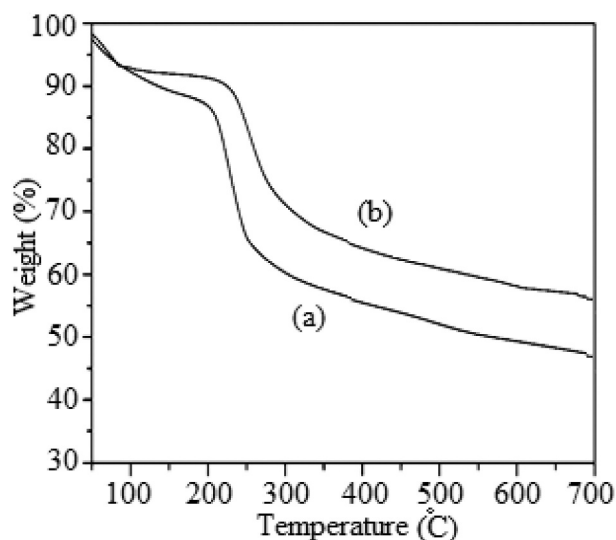


Figure 9. TGA curves of Cs (a) and CZn (b).

higher than that for Cs (228 °C). Cs and CZn exhibit temperatures at maximum decomposition of 238 °C and 260 °C, respectively. It is obvious that ZnO NPs slowdown the degradation rate of Cs which could be owing to the formation of physical bond between Cs and ZnO NPs in CZn which enhanced the overall strength of bonds in CZn. This in turn results in enhanced thermal stability of CZn compared to Cs. The residual mass percentages, resulting from Cs and CZn decomposition at 700°C, were 49% and 58%, respectively. These data show that by increasing temperature, CZn possessed high percentage of residual mass compare to Cs, owing to ZnO NPs existence in CZn.

3.2. Characterization of polypropylene – chitosan/ ZnO nanocomposite (PP/CZn)

3.2.1. FT-IR spectroscopy

Figure 10 represents the FT-IR spectra of PP, CZn, PP/10CZn and PP/20CZn. Figure 10a indicates that PP possesses bands at 2910, 1368 and 1445 cm^{-1} which

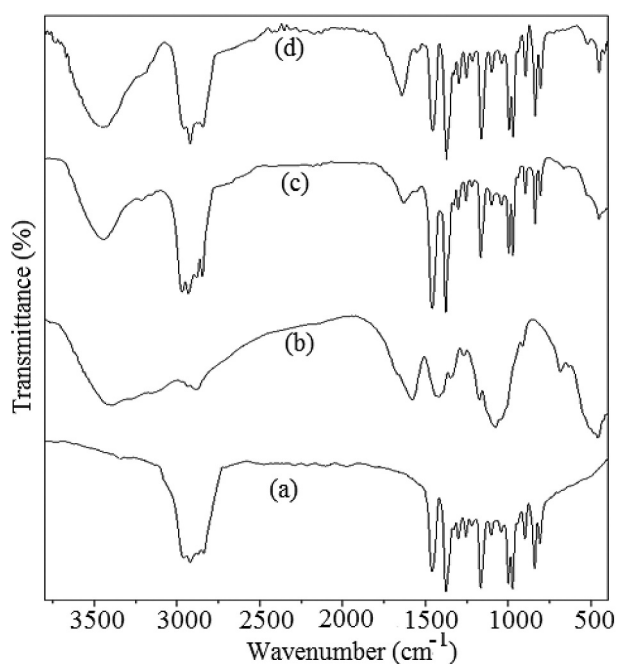


Figure 10. FT-IR spectra of PP (a), CZn (b), PP/10CZn (c) and PP/20CZn (d).

correspond to the stretching vibration of – CH group, bending of CH_3 and CH_2 , respectively. The bands at 1158, 988 and 835 cm^{-1} are attributed to C-C stretching vibration, C-H bending and C- CH_3 stretching vibration, respectively.^[39] As indicated from Figures 10c and 10d, PP/10CZn and PP/20CZn possess the characteristic peaks of both PP and CZn. However, the peaks attributed to stretching vibrations of amine and hydroxyl groups in CZn were shifted to higher wave numbers at 3431 cm^{-1} and 3442 cm^{-1} , respectively. This shift to higher wave number could be attributed to the good dispersion of CZn in PP matrix.^[40]

3.2.2. XRD

Figure 11 displays the XRD patterns of PP, CZn, PP/10CZn and PP/20CZn. As presented in Figure 11a, PP possesses peaks at 14.11°, 16.69°, 18.29°, 21.91°, 25.32°, 28.48° and 42.58°. The XRD pattern of CZn is shown in Figure 11b. From Figure 11c and 11d, PP/10CZn and PP/20CZn have the characteristic peaks of PP and CZn. The absence of some peaks corresponding to CZn in PP/CZn indicates that CZn as filler has no significant impact on overall crystallinity of PP in PP/CZn.

3.3. Effect of CZn on the biodegradability and mechanical properties of polypropylene

3.3.1. Biodegradation studies

The biodegradation behavior of PP and its nanocomposites was investigated by soil burial test. Figure 12a

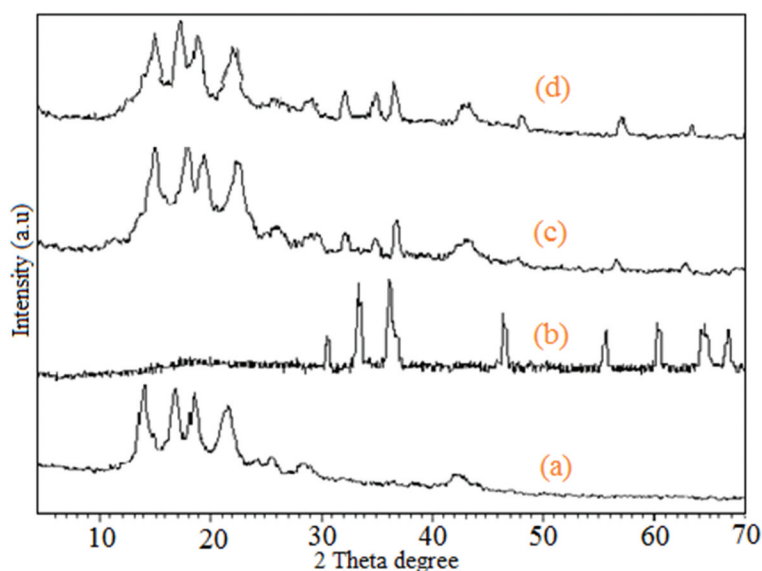


Figure 11. XRD patterns of PP (a), CZn (b), PP/10CZn (c) and PP/20CZn (d).

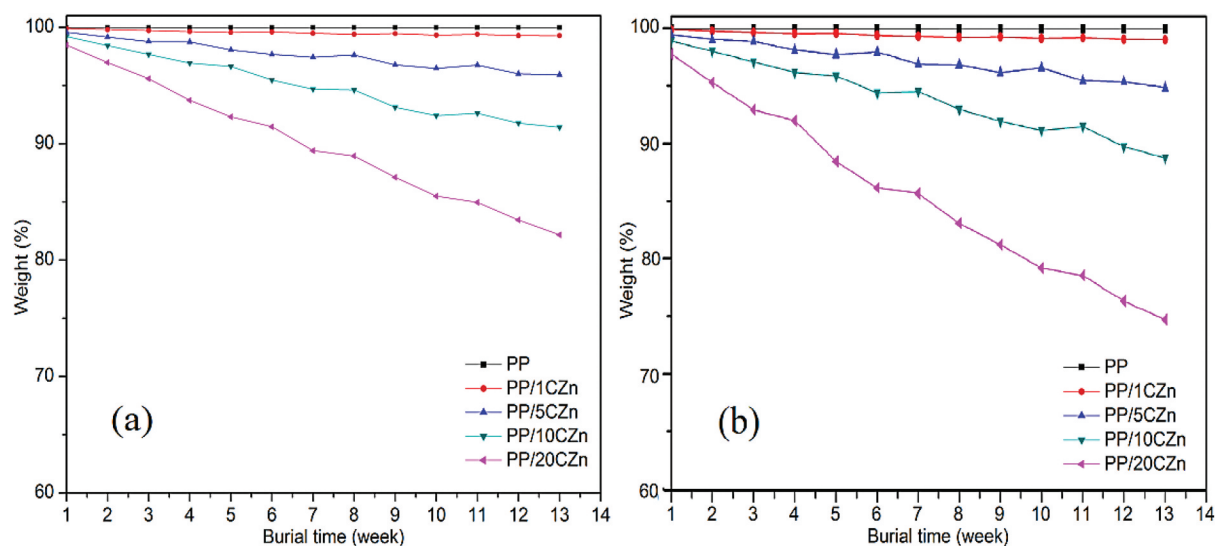


Figure 12. Weight loss curves of PP and PP/CZn after soil burial for 3 months (a) and UV pretreated PP and PP/CZP after soil burial (b).

displays the weight loss curves of PP, PP/1CZn, PP/5CZn, PP/10CZn and PP/20CZn as function of burial time in soil for three months. PP possessed almost no weight loss after burial in soil for three months due to the non-biodegradability of PP.^[41] PP nanocomposites loaded with CZn exhibited weight loss after soil burial for 3 months. The weight loss values for PP/1CZn, PP/5CZn, PP/10CZn and PP/20CZn after 3 months of soil burial were 0.72%, 4.07%, 8.59% and 17.82%, respectively. These results indicate that the gradual increase in CZn content led to increased values of weight loss. PP/20CZn is composed of 80% PP as matrix and 20% CZn as filler. Thus, when PP/CZn exhibits weight loss, this will affect the percentage of PP and CZn present

in the nanocomposite. PP/20CZn exhibits 17.82% weight loss after 3 months of soil burial. With respect to the composition of PP/20CZn, the weight loss coming from PP and CZn are about 14% and 3.8%, respectively. The addition of CZn to PP matrix has improved the biodegradation rate of PP which may be due to the effect of CZn on the segmental mobility of PP, accelerating the biodegradation process.^[42] PP/20CZn after soil burial exhibits weight loss due to the presence of CZn which is composed of Cs and ZnO NPs. The presence of CZn in PP/CZn has caused improved biodegradability coming from the increased hydrophilicity of PP/CZn. This in turn improves the attachment of microorganisms to surface of the nanocomposite

which utilizes the nanocomposite as carbon source. The growth of microorganisms on surface of nanocomposite results in bio deterioration of nanocomposite surface and bio fragmentation of the bonds present in nanocomposite. The cleavage of the main chains in nanocomposite leads to formation of low molecular weight fragments. Then, mineralization process occurs which leads to production of CO_2 and biomass.

The influence of UV pretreatment on the biodegradation rate of PP and its nanocomposites with CZn was studied through exposure of samples to UV radiation for 72 hours prior to soil burial. Figure 12b represents weight loss curves of UV pretreated PP, PP/1CZn, PP/5CZn, PP/10CZn and PP/20CZn as function of soil burial time. Regarding the UV pretreated PP; it exhibited no weight loss owing to its non-biodegradability. The UV pretreated PP/CZn possessed higher weight loss values compared to that of PP/CZn. The weight loss values of UV pretreated PP/1CZn, PP/5CZn, PP/10CZn and PP/20CZn after 3 months of soil burial were 0.98%, 5.14%, 11.24% and 25.29%, respectively. The obtained data indicated the significant impact of UV radiation on biodegradation rate of PP/CZn. The pretreatment of PP/CZn via exposure to UV radiation prior to soil burial has significant impact on the weight loss values. The presence of CZn acts as photocatalyst in the existence of UV radiation which initiates the oxidation of carbon chains present in the nanocomposite. This photooxidation of nanocomposite results in improved chain scission of bonds in nanocomposite. Also, it leads to incorporation of oxygen containing groups in the nanocomposite chains. Consequently, various functional groups like carbonyl, carboxylic acids, aldehydes, esters and alcohols are formed which accelerate the degradation process. In addition, the molecular weight of nanocomposite is significantly reduced to lower levels via oxidation. The oxidation products can be then biodegraded by microorganisms and converted into CO_2 , H_2O and biomass.^[43,44]

FT-IR was used as follow up tool to monitor the biodegradation process of PP/CZn. The FT-IR spectra of PP/20CZn before soil burial, after soil burial for 3 months without UV pretreatment and after soil burial for 3 months with UV pretreatment are displayed in Figure 13. The FT-IR spectrum of PP/20CZn after soil burial possessed the same characteristic peaks of PP/20CZn before soil burial but with lesser intensities. This shows the improved biodegradation rate of PP by addition of CZn. Additionally, the UV pretreated PP/20CZn exhibited absorption peaks with lesser intensities compared to that of the non-pretreated samples. This

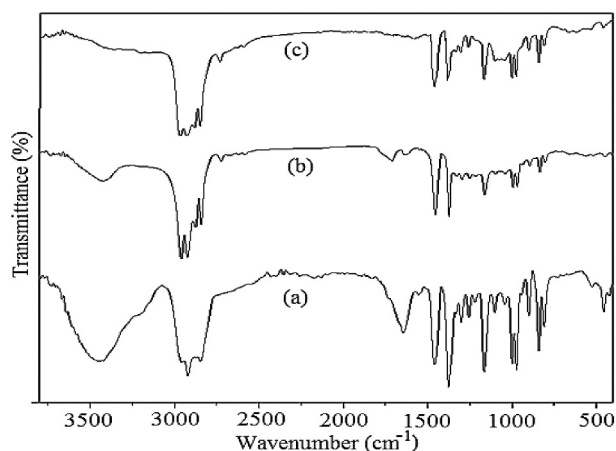


Figure 13. FT-IR spectra of PP/20CZn: before soil burial (a), after soil burial without pretreatment (b), after soil burial with UV pretreatment (c).

indicates that the biodegradation rate has been improved via the UV pretreatment of samples before soil burial.^[45]

3.3.2. Mechanical properties

The tensile test was used to study the mechanical behavior of PP, PP/1CZn, PP/5CZn and PP/10CZn films and the obtained data are presented in Figure 14. It is clear that the mechanical properties of PP exhibit remarkable change upon adding CZn. The tensile strength of PP is 30.69 MPa, while this value by incorporating 1% CZn into PP matrix (PP/1CZn) has increased to 31.35 MPa. The explanation of this behavior can be the improved interfacial bonding between filler (CZn) and matrix (PP), allowing stress transfer between and matrix and filler, causing an enhancement in tensile strength. The tensile strength has decreased to 27.68 MPa and 22.59 MPa by further increase in CZn content to 5% (PP/5CZn) and 10% (PP/10CZn), respectively. The decrease in tensile strength upon increase in CZn content can be attributed to agglomeration of CZn the polymer matrix, resulting in reduced contact area between PP and CZn as well as defect formation in the nanocomposites. Therefore, tensile strength is reduced.^[46] The elongation at break of PP is 21.21%, however this value has increased to 22.03% upon addition of 1% CZn to PP matrix (PP/1CZn). This can be owing to the improved interfacial bonding between PP and CZn. Also, CZn are uniformly dispersed into PP matrix. The elongation at break has decreased to 17.94% and 16.19% by further increase in CZn content up to 5% and 10%, respectively. The increased CZn content led to decrease in elongation at break which resulted from agglomeration of CZn, leading to phase separation and CZn defects in PP matrix.^[47]

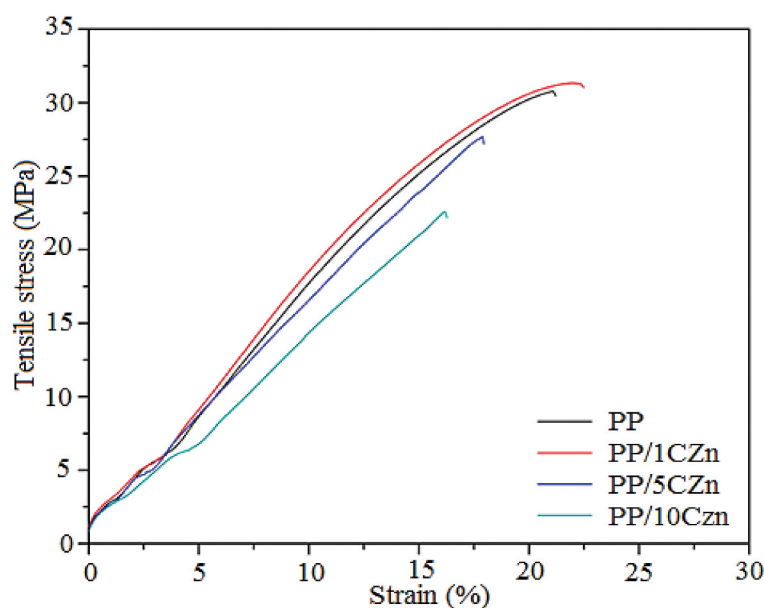


Figure 14. Tensile stress-strain curves of PP and PP/CZn.

4. Conclusion

Chitosan/ZnO nanocomposite (CZn) was successfully prepared by solution mixing via the electrostatic repulsion between the positively charged chitosan (Cs) chains and ZnO NPs surface. Polypropylene – chitosan/ZnO nanocomposites (PP/CZn) loaded with various amounts of CZn have been prepared by melt processing. PP/1CZn exhibited improved tensile strength and elongation at break relative to PP, while the further increase in CZn content up to 5% and 10% has resulted in decreased tensile strength and elongation at break. PP/CZn possessed improved biodegradability compared to PP. After 3 months of soil burial, PP showed no weight loss,; however, PP/1CZn, PP/5CZn, PP/10CZn and PP/20CZn revealed weight loss values of 0.72%, 4.07%, 8.59% and 17.82%, respectively. The UV pretreated PP/CZn exhibited improved biodegradability compared to untreated samples, showing that UV radiation exhibited remarkable effect on biodegradation rate of PP/CZn.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Notes on contributors

Mohamed S. A. Darwish is an associate professor at polymer laboratory, petrochemicals department, Egyptian Petroleum Research Institute.

Mohamed H. Mostafa is a research assistant at polymer laboratory, petrochemicals department, Egyptian Petroleum Research Institute.

Labiba I. Hussein is a professor of petrochemicals at polymer laboratory, petrochemicals department, Egyptian Petroleum Research Institute.

Abdaleem H. Abdaleem is a professor of chemistry, faculty of science, Menofia University.

Moataz A. Elsayy is an assistant professor at polymer laboratory, petrochemicals department, Egyptian Petroleum research institute.

ORCID

Mohamed S. A. Darwish  <http://orcid.org/0000-0002-4598-8847>

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